Benchmarking Subset Aggregation for Classical Causal Discovery Under Marginalization Error

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Abstract

Subset-estimate-aggregate pipelines are attractive for causal discovery because they recycle small local structure-learning problems into one global graph, but their soundness is fragile when local graphs are learned on observed-variable subsets of a larger system. We study this issue in a fixed-bank benchmark rather than proposing a new discovery algorithm. Across 48 synthetic datasets spanning Erdős–Rényi and scale-free graphs, four structural regimes, two sample sizes, and three seeds, we keep one bank of 40 local CPDAGs per dataset fixed and vary only weighting and merge rules. The benchmark first audits subset distortion against induced-subgraph truth: non-adjacency claims are usually valid (0.980–0.993 by regime), but direction claims are rarely valid (0.110–0.158) and are abstained on 0.888–0.900 of opportunities. Under this distortion, compatibility weighting yields only a small hard-regime median reduction in false-orientation rate over matched uniform weighting (0.0286 for CompatExp+ DetThreshold), far below the pre-registered 0.10 target and statistically null ($p = 0.351$). Full-graph baselines remain much stronger: GES reaches mean SHD 6.83 and mean orientation precision 0.748, whereas the best subset variants remain near SHD 18.4–18.5 and orientation precision 0.417–0.426. The main conclusion is negative but useful: in this low-compute setting, subset compatibility is not enough to overcome marginalization error, and benchmark design choices such as subset size matter more than the proposed weighting heuristic.

1 Introduction

Many recent causal discovery papers use internal agreement signals to select, evaluate, or aggregate candidate graphs without access to ground truth [Viinikka et al., 2018, Faller et al., 2024, Schkoda et al., 2025]. A natural extension is to learn local graphs on overlapping subsets of variables and then merge their symbolic claims into one global CPDAG. That pattern already appears in subset-estimate-aggregate pipelines [Wu et al., 2025], in bagging-style graph aggregation [Wang and Peng, 2014, Debeire et al., 2024], and in newer consistency-guided aggregation work [Ninad et al., 2025]. The unresolved issue is not whether graph aggregation can help in general, but whether compatibility-style weighting is actually useful when local graphs come from arbitrary observed-variable subsets of one larger system.

That setting is methodologically dangerous. A CPDAG learned on a subset of observed variables need not match the CPDAG of the induced subgraph on that subset. Omitted observed variables can create effective confounding, alter collider patterns, and break the transport of local orientations. If so, disagreement among local graphs may reflect structural distortion rather than finite-sample noise. Our paper therefore studies subset aggregation as an empirical benchmark with an explicit soundness audit, not as a new causal discovery principle.

We build on the proposal in this workspace and ask a narrower question: when one fixed bank of local subset CPDAGs is held constant, do compatibility-based weights improve global recovery relative to uniform weighting and a non-falsification bootstrap-stability control? The answer in our

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benchmark is mostly no. The subset audit shows that local non-adjacency statements are usually correct, but adjacency, direction, and collider claims are heavily distorted. Under these conditions, compatibility weighting produces only a 0.0286 hard-regime median reduction in false-orientation rate over matched uniform weighting, with a paired Wilcoxon p-value of 0.351. At the same time, strong full-graph baselines such as GES and PC-Stable remain far better than every subset variant we tested.

Our contributions are:

- We define a reproducible fixed-bank benchmark for overlapping-subset CPDAG aggregation under explicit marginalization error, covering 48 synthetic datasets with CPU-only execution and shared local evidence across all subset methods.
- We audit subset validity before interpreting aggregation gains. Across regimes, direction validity is only 0.110–0.158 and collider validity only 0.088–0.140, even though non-adjacency validity remains 0.980–0.993.
- We show that compatibility weighting is not a robust remedy in this setting: its observed gains over uniform weighting are small, statistically null, and substantially weaker than improvements obtained by changing subset size or using full-graph baselines.

Section 2 situates the benchmark in prior work on internal validation and graph aggregation. Section 3 defines the fixed-bank aggregation setup and the subset-validity audit. Section 4 presents the benchmark, main results, and ablations. Section 5 discusses implications and limitations, and Section 6 concludes.

2 Related Work

Internal validation for causal discovery. Agreement-based evaluation without ground truth predates recent causal discovery benchmarks. Intersection-Validation [Viinikka et al., 2018] uses overlap agreement to evaluate structure learning, while Self-Compatibility [Faller et al., 2024] turns subset disagreement into an internal reliability signal. Leave-One-Variable-Out extends this line toward model selection rather than aggregation [Schkoda et al., 2025]. Our benchmark borrows the idea that disagreement can be informative, but tests it in a harder setting where local graphs are learned on fixed observed-variable subsets and then merged.

Subset aggregation and consistency-guided merging. SEA shows that sample-estimate-aggregate pipelines can be effective at larger scale with learned aggregation modules [Wu et al., 2025]. Ninad et al. study consistency-guided aggregation for vector-valued variables and explicitly analyze failure modes when aggregation maps are invalid [Ninad et al., 2025]. Those works substantially narrow novelty for any new subset-aggregation method. Our contribution is therefore empirical and narrower: a controlled benchmark that isolates weighting and merging decisions while auditing the local evidence for marginalization-induced distortion.

Classical graph aggregation and competing full-graph baselines. Graph bagging is longstanding in causal structure learning. DAGBag aggregates bootstrap DAGs to reduce finite-sample false positives [Wang and Peng, 2014], and similar ideas appear in time-series causal discovery [Debeire et al., 2024]. We also compare against standard full-graph learners and selectors: PC-Stable [Colombo and Maathuis, 2014], GES [Chickering, 2002], NOTEARS [Zheng et al., 2018], and self-compatibility-based model selection [Faller et al., 2024]. AutoML-style baseline selection is another alternative to local aggregation [Chan et al., 2024], and Malinsky [2024] argues for conservatism in constraint-based model selection. Unlike these approaches, our benchmark fixes the local subset evidence and asks only whether compatibility-based weighting adds value once marginalization error is already present.

3 Method

3.1 Fixed-Bank Benchmark Setup

This paper does not propose a new discovery algorithm. The object of study is a family of symbolic aggregators that all operate on the same precomputed bank of local CPDAGs. For each dataset,

the benchmark records 20 overlapping subsets of size $k = 8$ and learns two bootstrap CPDAGs per subset using PC-Stable with $\alpha = 0.01$, yielding 40 local graphs per dataset. The executed environment used dataset seeds $\{11, 23, 37\}$, bootstrap seeds $\{101, 131\}$, subset-bank seed 509, two CPU workers, and no GPUs.

All subset methods therefore share identical local evidence. Differences between methods come only from how local graphs are weighted and how weighted claims are merged. This isolates the contribution of compatibility weighting from confounds due to subset construction, local learning, or stochastic variation in the evidence bank.

3.2 Claim Extraction

From each local graph g on subset S_g, we extract only explicit symbolic claims:

$$\texttt{adj}(i,j), \quad \texttt{nonadj}(i,j), \quad \texttt{dir}(i \to j), \quad \texttt{coll}(i \to k \leftarrow j).$$

Undirected edges contribute adjacency evidence only. Missing orientations are treated as abstentions rather than contradictions.

3.3 Compatibility Weighting

For two local graphs g_a and g_b with overlap $O = S_a \cap S_b$, we compare only claims whose variables lie in O. Contradictions are counted only when both graphs make explicit opposing statements, namely adjacency versus non-adjacency, opposite directions on the same edge, orientation versus non-adjacency, or incompatible collider statements on the same triple. Let $\mathrm{contr}(g_a, g_b)$ denote the number of explicit contradictions and $\mathrm{comp}(g_a, g_b)$ the number of comparable claims. We define the overlap contradiction rate

$$r(g_a, g_b) = \begin{cases} \mathrm{contr}(g_a, g_b)/\mathrm{comp}(g_a, g_b), & \mathrm{comp}(g_a, g_b) > 0, \\ 0, & \mathrm{comp}(g_a, g_b) = 0. \end{cases}$$

For each local graph g, the compatibility score is the mean contradiction rate against overlapping neighbors,

$$C_g = \frac{1}{|\mathcal{N}(g)|} \sum_{h \in \mathcal{N}(g)} r(g, h).$$

Our primary weighting rule is

$$w_g = \exp(-2C_g).$$

We compare this CompatExp rule against uniform weighting and a bootstrap-stability control that scores each subset using agreement between its two bootstrap graphs. In the main benchmark, the median effective sample size proxy is 40.0 for uniform weighting and 39.40 for CompatExp, while median weighted opposition falls from 4.0 to 3.056.

3.4 Merge Rules

Aggregators combine weighted support tables and then apply deterministic consistency checks to produce a final CPDAG. The main paper studies two merge rules.

Wilson-style screen. For a binary claim family, let

$$\hat{p} = \frac{\text{support}}{\text{support} + \text{opposition}}, \qquad n_{\text{eff}} = \frac{(\sum_g w_g)^2}{\sum_g w_g^2}.$$

A claim is accepted if the one-sided 90% Wilson lower bound exceeds 0.5. Because local graphs overlap heavily, n_{eff} is only a shrinkage heuristic and not a calibrated sample size.

Deterministic thresholding. DetThreshold accepts a claim whenever the weighted support ratio exceeds 0.6. We also study deterministic rank-based ablations, but our main comparisons use Wilson and DetThreshold because both were pre-registered across all 48 datasets.

Table 1: Subset-validity audit against induced-subgraph truth. Each number is the regime-level validity or abstention rate computed from local graphs before aggregation. High non-adjacency validity contrasts with poor orientation validity, showing that the local bank is structurally distorted in exactly the claim families subset aggregation most needs.

Regime	Adj val.	Nonadj val.	Dir val.	Coll. val.	Dir abst.
Linear Gaussian	0.5094	0.9932	0.1537	0.1319	0.8881
Mild misspecification	0.5156	0.9883	0.1473	0.1275	0.8904
Near-unfaithful linear	0.4811	0.9798	0.1585	0.1404	0.8987
Nonlinear ANM	0.4906	0.9815	0.1101	0.0882	0.9004

3.5 Subset-Validity Audit

The benchmark interprets aggregation only after auditing the local evidence against induced-subgraph truth. For every sampled subset S, we compare each learned local graph against the true CPDAG of the induced subgraph on S. This yields per-claim validity, contradiction, and abstention rates before any aggregation takes place. The audit is essential because compatibility weighting can only be meaningful if local disagreement is at least partially tracking finite-sample error rather than mostly reflecting subset-induced distortion.

4 Experiments

4.1 Benchmark and Evaluation Protocol

The executed benchmark contains 48 datasets from the Cartesian product of two graph families (Erdős–Rényi and scale-free), four regimes (linear Gaussian, nonlinear ANM, near-unfaithful linear, and mild misspecification), two sample sizes ($n = 200$ and $n = 1000$), and three seeds. Every dataset has $p = 20$ observed variables. The benchmark manifest confirms that true CPDAG edge counts range from 15 to 23 in the Erdős–Rényi family and are 19 in every scale-free instance.

We evaluate five full-graph baselines (PC, GES, NOTEARS-L1, DAGBag-PC, and SC-Select) and six fixed-bank subset methods (Uniform, BootstrapStability, and CompatExp combined with Wilson or DetThreshold merging). Metrics include SHD, skeleton F1, orientation precision, false-orientation rate, fraction of recovered adjacencies left undirected, runtime, and peak memory. The environment ran with Python $3.12.3$, two CPU workers, and no GPUs; the timing pilot projected total benchmark cost at 0.206 hours and did not trigger any fallback reductions.

4.2 Subset Audit: Local Evidence Is Structurally Distorted

Table 1 summarizes the subset-validity audit. The picture is asymmetric. Non-adjacency claims are highly reliable in every regime, with validity between 0.9798 and 0.9932. By contrast, adjacency validity stays near chance at 0.4811–0.5156, direction validity is only 0.1101–0.1585, and collider validity is only 0.0882–0.1404. The local bank is also highly abstaining on orientation-sensitive claims: direction abstention is 0.8881–0.9004 and collider abstention is 0.9853–0.9875.

These numbers sharply constrain what aggregation can plausibly recover. If local direction claims are mostly wrong or absent, compatibility weighting has little trustworthy orientation signal to amplify. Figure 2 makes the same point at the dataset level: the correlation between direction validity and the CompatExp-minus-Uniform change in false-orientation rate is weak (Pearson -0.066, Spearman -0.218).

4.3 Main Results: Compatibility Weighting Does Not Rescue Subset Aggregation

Table 2 reports the main benchmark summary. Among subset methods, the differences are minimal. Uniform+DetThreshold achieves mean SHD 18.38, mean skeleton F1 0.7325, mean orientation precision 0.4264, and mean false-orientation rate 0.5736. The corresponding CompatExp+DetThreshold values are SHD 18.52, skeleton F1 0.7307, orientation precision 0.4173, and false-orientation rate 0.5827. The Wilson variants are nearly identical to their deterministic counterparts.

Table 2: Main benchmark summary over 48 datasets. Best values are in **bold**. Higher is better for skeleton F1 and orientation precision; lower is better for SHD, false-orientation rate, and runtime. The subset methods are tightly clustered, while full-graph baselines dominate them on every structural metric.

Method	SHD ↓	Skeleton F1 ↑	Orient. Prec. ↑	False-orient. ↓	Runtime (s) ↓
PC	7.29	0.8967	0.7239	0.2761	**0.3409**
GES	**6.83**	**0.9047**	**0.7477**	**0.2523**	11.9116
NOTEARS-L1	18.83	0.0000	0.0000	0.0000	0.5132
DAGBag-PC	8.10	0.8839	0.7008	0.2992	2.8400
SC-Select	8.04	0.8840	0.6832	0.3168	1.4722
Uniform+Wilson	18.38	0.7325	0.4264	0.5736	**0.0100**
Uniform+DetThreshold	18.38	0.7325	0.4264	0.5736	0.0139
BootstrapStability+Wilson	18.46	0.7313	0.4179	0.5821	0.0163
BootstrapStability+DetThreshold	18.48	0.7309	0.4175	0.5825	0.0165
CompatExp+Wilson	18.50	0.7309	0.4173	0.5827	0.0261
CompatExp+DetThreshold	18.52	0.7307	0.4173	0.5827	0.0261

The pre-registered hard-regime test leads to the same conclusion. On the near-unfaithful and mild-misspecification slice, CompatExp+DetThreshold reduces the median false-orientation rate by only 0.0286 relative to Uniform+DetThreshold. That effect falls well short of the pre-registered 0.10 success criterion and is statistically null (paired Wilcoxon $p = 0.351$, median paired difference 0.0, bootstrap CI $[0.0, 0.0]$). Against BootstrapStability+DetThreshold, the difference is again 0.0 with $p = 0.638$.

More importantly, all subset variants are substantially worse than strong full-graph baselines. GES is the best overall method with mean SHD 6.83, mean skeleton F1 0.9047, mean orientation precision 0.7477, and mean false-orientation rate 0.2523. PC and DAGBag-PC are close behind. Relative to CompatExp+DetThreshold, the paired median false-orientation rate difference is 0.2752 for GES, 0.2638 for PC, and 0.2344 for DAGBag-PC, all with $p < 10^{-7}$.

4.4 Ablations: Subset Size Matters More Than Compatibility Weighting

Figure 3 and the ablation summaries reinforce the negative main result.

Alternative compatibility weights. On the hard-slice ablation, CompatRank+DetThreshold reaches median SHD 17.0 and median false-orientation rate 0.6583, while CompatTopHalf+DetThreshold reaches median SHD 17.5 and median false-orientation rate 0.7450. Neither variant improves on the main CompatExp rule in a meaningful way.

Claim-family ablations. Removing collider claims is mostly neutral: CompatExp+DetThreshold+NoColliders yields median SHD 16.5 and median false-orientation rate 0.6049. By contrast, removing directional information is destructive. The adjacency-only version has median SHD 20.5, mean orientation precision 0.1279, mean false-orientation rate 0.7888, and fraction undirected 0.4893. In other words, the subset methods still need orientation claims even though those claims are noisy.

Contradiction definition. Treating one-sided abstention as weak disagreement modestly helps on the hard slice, improving mean orientation precision from 0.4173 in the main method to 0.4399 and lowering mean false-orientation rate from 0.5827 to 0.5601. This suggests that the explicit-claims-only contradiction rule is not uniquely responsible for the observed behavior.

Subset-size sensitivity. Subset size has the largest effect of any ablation. At $k = 10$, Uniform+DetThreshold improves to mean SHD 12.42, mean skeleton F1 0.8051, mean orientation precision 0.5497, and mean false-orientation rate 0.4503; CompatExp+DetThreshold is nearly identical with SHD 12.54 and false-orientation rate 0.4524. At $k = 6$, both methods deteriorate sharply: Uniform+DetThreshold has mean SHD 21.63 and mean false-orientation rate 0.6302, while CompatExp+DetThreshold has mean SHD 21.96 and mean false-orientation rate 0.6412. This is a central result of the paper: subset design choices dominate the weighting choice.

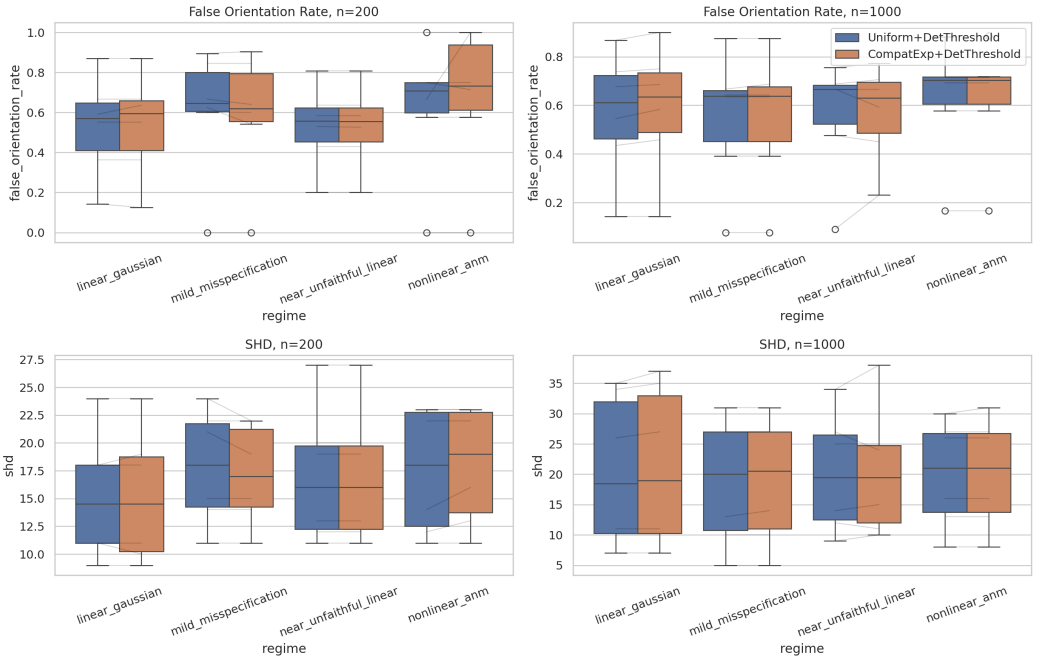

Figure 1: Distribution of false-orientation rates across regimes and sample sizes. The paired links between Uniform+DetThreshold and CompatExp+DetThreshold show that most dataset-level differences are small.

5 Discussion and Limitations

The benchmark supports a constrained conclusion. Compatibility weighting is not harmful in the sense of catastrophic failure, but it is also not a reliable solution to the subset-marginalization problem. The local evidence bank already contains severe orientation distortion: direction validity is below 0.16 in every regime, collider validity below 0.15, and abstention on those claim families is extremely high. Under those conditions, the weighted merger mostly shuffles weak or corrupted orientation evidence rather than extracting robust global signal.

The strongest practical lesson is that benchmark geometry matters more than the proposed weighting heuristic. Moving from $k = 8$ to $k = 10$ improves both uniform and compatibility-weighted subset methods by roughly six SHD points and lowers mean false-orientation rate by about 0.12–0.13, whereas switching from uniform to compatibility weighting at fixed $k = 8$ barely changes the outcome. This suggests that future work on subset aggregation should prioritize subset construction, local learner fidelity, and marginalization-aware claim selection over increasingly elaborate weighting rules.

The paper also has several limitations. First, it is a synthetic benchmark only; real data are intentionally out of scope because they would weaken the ground-truth claims central to the soundness audit. Second, the subset methods all use PC-Stable local learners and symbolic merging, so the conclusions do not automatically transfer to learned aggregators such as SEA. Third, the Wilson screen uses an overlap-sensitive effective sample size heuristic rather than a calibrated uncertainty estimate. Fourth, the planned LOVO baseline was not run because the scoring recipe was not operationally unambiguous without adding unregistered design choices. Finally, the benchmark fixes $p = 20$ and a low-compute CPU setting, so larger problems may show different trade-offs.

6 Conclusion

We presented a fixed-bank benchmark for subset aggregation in classical causal discovery under explicit marginalization error. The benchmark first audits whether local subset claims are trustworthy

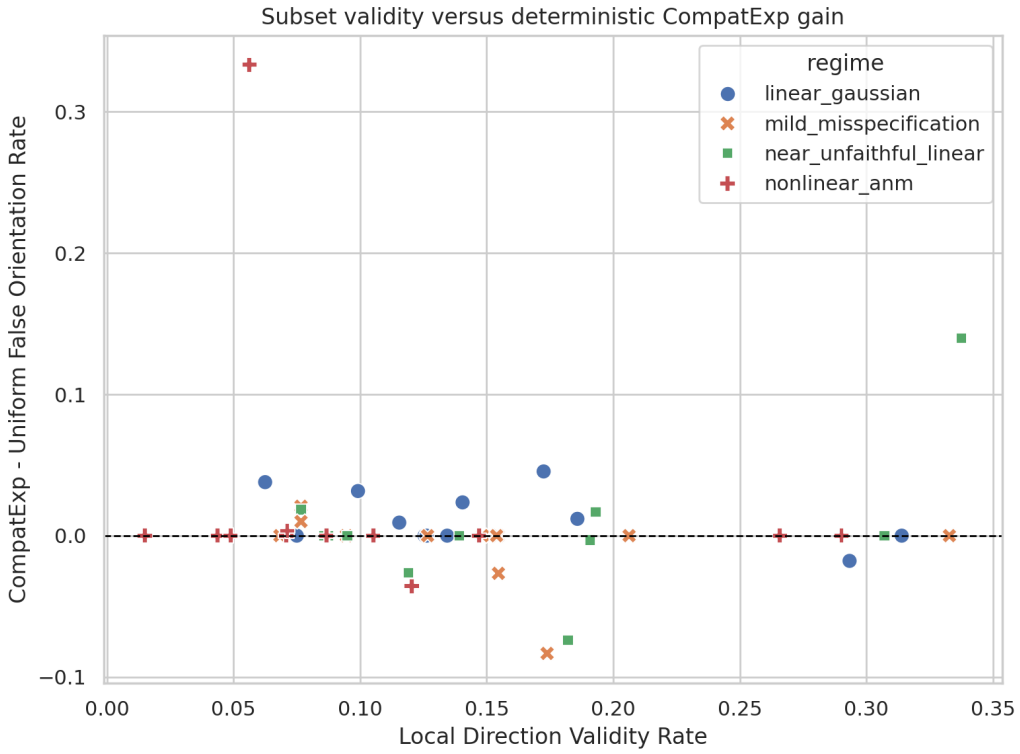

Figure 2: Subset direction-validity rate versus the dataset-level change in false-orientation rate from replacing Uniform+DetThreshold with CompatExp+DetThreshold. The weak negative correlations (Pearson -0.066, Spearman -0.218) indicate no strong evidence that compatibility weighting becomes useful exactly when local orientation validity improves.

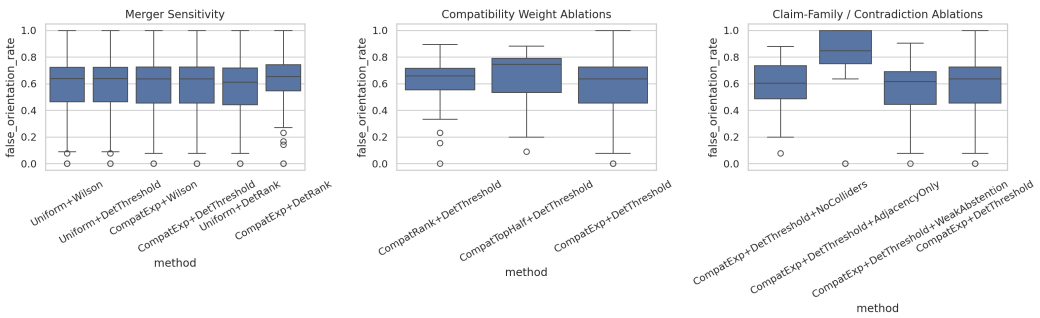

Figure 3: Merger and ablation summary. The broad pattern is stable: changing the merge rule or compatibility variant does little, whereas subset size and the availability of directional claims move performance substantially more.

relative to induced-subgraph truth and then asks whether compatibility weighting improves global recovery once the local evidence bank is held fixed.

The main findings are negative but informative. Local non-adjacency claims are usually valid (0.980–0.993), but local direction claims are rarely valid (0.110–0.158) and highly abstaining. In that regime, compatibility weighting produces only a 0.0286 hard-regime median reduction in false-orientation rate over uniform weighting, far below the pre-registered success threshold and statistically null. Meanwhile, full-graph baselines such as GES, PC, and DAGBag-PC remain much

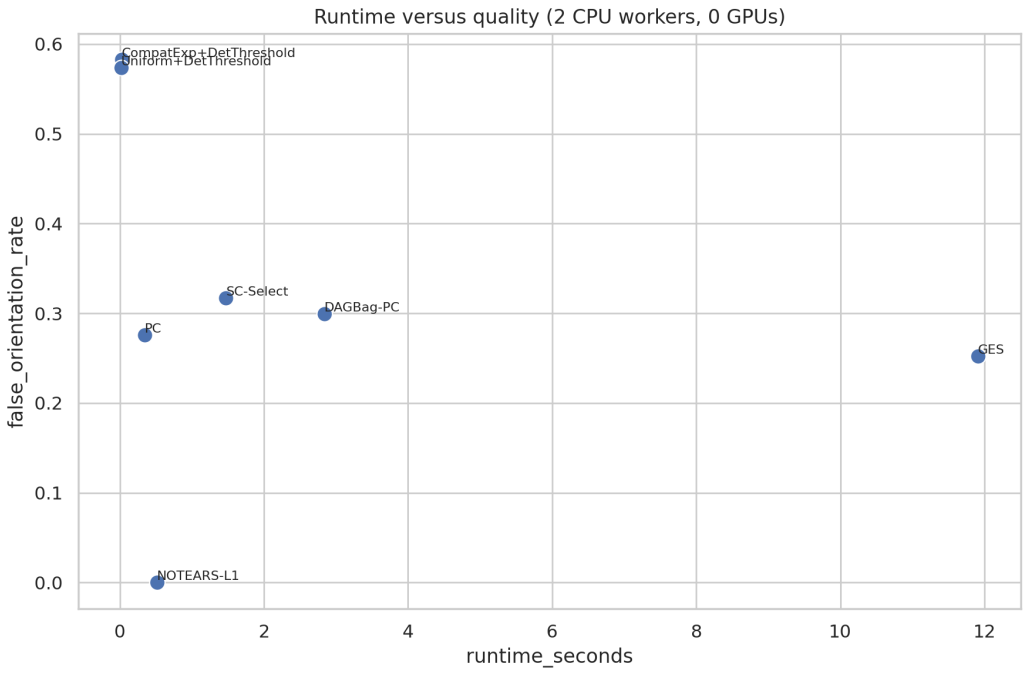

Figure 4: Runtime versus quality trade-off. Subset aggregation is very fast at merge time, but its structural quality remains far behind the strongest full-graph baselines. All reported runs used two CPU workers and zero GPUs.

stronger, with mean SHD between 6.83 and 8.10 versus approximately 18.4–18.5 for the subset methods.

The broader implication is that overlap compatibility is not a substitute for valid local evidence. In low-compute fixed-variable subset aggregation, marginalization error appears to be the dominant bottleneck, and benchmark choices such as subset size matter more than the weighting heuristic itself. Future work should focus on marginalization-aware local representations, better subset design, or aggregation rules that can explicitly reason about omitted-variable distortion instead of treating local CPDAGs as directly transportable evidence.

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